

A Global Quadratic Sard Theorem for maps from a Hilbert space to the plane

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Status. Candidate partial result, likely valid, subject to human review.

Source question. Lerario, Rizzi, and Tiberio define a class $\mathcal{P}_d^m(H)$ of polynomial maps from a real Hilbert space to $\mathbb{R}^m$, with a weakly continuous and locally Lipschitz differential. After constructing counterexamples in degrees at least three, their Remark 3 says:

The case $d = 2$ is special. The authors expect every $f \in \mathcal{P}_2^m(H)$ to satisfy the Sard property globally on H .

The exact source passage is on page 3 of arXiv:2407.02296v4 and is reproduced below.

Remark 2. The constant $\beta_0(d, m)$ of Theorem A comes from semialgebraic geometry, and its origin is discussed in Section 1.2. We sometimes call it the *semialgebraic constant*.

By building upon Kupka's counterexamples [Kup65], we prove that the quantitative assumption $q > \beta_0$ in Theorem A cannot be dispensed of. The following result corresponds to Theorem 39. We denote by $B_X(r)$ the closed ball of radius r in a Banach space X .

Theorem B (Counterexamples to Sard). *Let $d, m \in \mathbb{N}$, with $d \geq 3$, and let $1 < q < (d-1)^{1/d}$. There exist a Hilbert space H , and $f \in \mathcal{P}_d^m(H)$ such that $K = \text{Crit}(f) \cap B_H(r)$ is compact for all $r > 0$, with*

$$\limsup_{n \rightarrow \infty} \Omega_n(K, H)^{1/n} \leq q^{-1},$$

and $f : H \rightarrow \mathbb{R}^m$ does not verify the Sard property, namely $\mu(f(\text{Crit}(f) \cap K)) > 0$. Therefore, the semialgebraic constant $\beta_0(d, m)$ of Theorem A satisfies

$$\beta_0(d, m) \geq (d-1)^{1/d}, \quad \forall m \in \mathbb{N}, d \geq 3.$$

Remark 3. The case $d = 2$, which is left out from Theorem B, is special. This is related to the fact that the complexity of semialgebraic sets defined by quadratic equations has a different behaviour compared to the case $d \geq 3$. We currently expect that *any* $f \in \mathcal{P}_2^m(H)$ satisfies the Sard property globally on H . This will be the subject of a future work.

The result here proves the expectation completely for codomain dimension two, with no commutativity or regularity assumption on the self-adjoint Hessian pencil.

1 Statement

Every $f \in \mathcal{P}_2^2(H)$ has a representation

$$f_j(x) = c_j + 2\langle a_j, x \rangle + \langle A_j x, x \rangle, \quad j = 1, 2, \tag{1}$$

where A_1, A_2 are compact self-adjoint operators. Put

$$K = \ker A_1 \cap \ker A_2, \quad L_K z = (\langle a_1, z \rangle, \langle a_2, z \rangle).$$

Theorem 1 (global quadratic Sard theorem in target dimension two). *Let H be a real Hilbert space. Every $f \in \mathcal{P}_2^2(H)$ satisfies the global Sard property:*

$$\mathcal{L}^2(f(\text{Crit}(f))) = 0.$$

2 Proof intuition

A critical point has a projective multiplier $[\lambda_1 : \lambda_2]$ for which the scalar quadratic $\lambda_1 f_1 + \lambda_2 f_2$ is stationary. On a bounded ball, all pairs consisting of a stationary point and its multiplier form a compact incidence set when the ball is given its weak topology.

The key observation compares two such pairs. If x, z have nearby multipliers λ, μ , the two exact quadratic Taylor formulas imply

$$(\lambda + \mu) \cdot (f(z) - f(x)) = \langle A_{\lambda - \mu}(z - x), z - x \rangle.$$

Consequently their critical values lie in a thin strip: its normal thickness is bounded by a constant times the distance between the multipliers.

The pencil can be singular in uncountably many directions, so uniqueness of the stationary point is false in general. The decisive replacement is that only countably many multiplier directions can have a stationary fiber on which f is nonconstant. Indeed, on an affine chart of the multiplier circle, write $B_t = A_2 - tA_1$ and $K_t = \ker B_t$. If the quadratic part varies on K_t , vectors from distinct K_t give a biorthogonal system. If only the linear part varies, the stationary gradients give another biorthogonal system. Every biorthogonal system in a separable Hilbert space is countable.

Away from those active directions, a multiplier has a unique critical *value*, even when it has many stationary points. Compactness of the incidence set makes that value uniformly continuous on compact multiplier sets. Splitting the multiplier circle into arcs of length δ covers the image by $O(\delta^{-1})$ rectangles, each with one side $O(\delta)$ and the other $o(1)$. Countably many active directions can be put inside open arcs of arbitrarily small total length. The strip estimate finishes the proof.

3 Proof

3.1 Quadratic representation and compact Hessians

For a scalar component of f , subtract its value and differential at the origin. Its restriction to every finite-dimensional subspace is a homogeneous quadratic polynomial. Polarization therefore gives a continuous symmetric bilinear form on H , represented by a bounded self-adjoint operator. This gives (1). Moreover,

$$Df_j(x)h = 2\langle A_j x + a_j, h \rangle.$$

If $x_n \rightharpoonup 0$, weak continuity of Df into the operator norm gives $\|A_j x_n\| \rightarrow 0$. The weak-null sequence characterization of compact operators on Hilbert space shows that A_j is compact.

Only a separable part of H matters. Indeed, the closed span of $a_1, a_2, \text{ran } A_1, \text{ran } A_2$ is separable, and f is constant on its orthogonal complement. We may consequently assume H is separable. On bounded subsets, each compact quadratic form $x \mapsto \langle A_j x, x \rangle$ is weakly continuous. Thus f is weakly continuous on bounded balls.

3.2 The common kernel

On K , the restriction of every derivative is the fixed linear map $2L_K$. If $\text{rank } L_K = 2$, every $Df(x)$ is onto and there are no critical points.

Suppose $\text{rank } L_K = 1$. Any multiplier $[\lambda]$ at a critical point must annihilate $L_K(K)$, so the projective multiplier is unique. Write

$$A_\lambda = \lambda_1 A_1 + \lambda_2 A_2, \quad a_\lambda = \lambda_1 a_1 + \lambda_2 a_2.$$

If x, z are critical, both solve $A_\lambda u + a_\lambda = 0$. Hence $A_\lambda(z - x) = 0$, and the exact Taylor formula gives

$$\lambda \cdot (f(z) - f(x)) = \langle A_\lambda(z - x), z - x \rangle = 0.$$

All critical values lie on one affine line and are null.

It remains to consider $L_K = 0$. The linear and quadratic terms vanish on K , including all cross terms, so f factors exactly through $K^\perp$. We replace H by $K^\perp$; henceforth $K = \{0\}$.

3.3 The multiplier incidence and the strip estimate

Let $P = \mathbb{RP}^1$ be the projective multiplier circle. For $R > 0$ define

$$\mathcal{I}_R = \{([\lambda], x) \in P \times B_H(R) : A_\lambda x + a_\lambda = 0\}.$$

The equation is invariant under changing the sign of a representative. Give $B_H(R)$ its weak topology. It is compact and metrizable. If $[\lambda_n] \rightarrow [\lambda]$ and $x_n \rightarrow x$, choose locally aligned unit representatives. Compactness of A_λ and norm continuity of the pencil give

$$A_{\lambda_n} x_n + a_{\lambda_n} \rightarrow A_\lambda x + a_\lambda \quad \text{in norm.}$$

Thus $\mathcal{I}_R$ is compact. Its image under $([\lambda], x) \mapsto f(x)$ is exactly $f(\text{Crit}(f) \cap B_H(R))$.

We need one quantitative identity. Take $([\lambda], x), ([\mu], z)$ in $\mathcal{I}_R$ and choose aligned unit representatives. Put $h = z - x$. Taylor expansion at the two stationary points gives

$$\begin{aligned} \lambda \cdot (f(z) - f(x)) &= \langle A_\lambda h, h \rangle, \\ \mu \cdot (f(z) - f(x)) &= -\langle A_\mu h, h \rangle. \end{aligned}$$

Adding them yields the exact identity

$$(\lambda + \mu) \cdot (f(z) - f(x)) = \langle A_{\lambda - \mu} h, h \rangle. \tag{2}$$

Since $\|h\| \leq 2R$ and $f(B_H(R))$ is bounded, (2) implies

$$|\lambda \cdot (f(z) - f(x))| \leq C_R \|\lambda - \mu\| \tag{3}$$

for a constant depending only on R, f . Thus values whose multipliers lie in one short projective arc of length ℓ lie in a strip of normal width at most $2C_R \ell$.

3.4 Only countably many stationary fibers are active

We use the following elementary separability fact.

Lemma 2 (biorthogonal families). *Let H be a separable Hilbert space. If $(u_\alpha, v_\alpha)_{\alpha \in I} \subset H \times H$ satisfies*

$$\langle u_\alpha, v_\beta \rangle = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

then I is countable.

Proof. Rescale so that $\|v_\alpha\| = 1$. For $n \geq 1$, the vectors v_α with $\|u_\alpha\| \leq n$ are pairwise $1/n$ -separated, because $\langle u_\alpha, v_\alpha - v_\beta \rangle = 1$. A separated subset of a separable metric space is countable, and the union over n is all of I . $\square$

Call a multiplier direction *active* if the stationary set for that direction is nonempty and its image under f contains more than one point.

Lemma 3 (countability of active directions). *The active projective multiplier directions form a countable subset of $\mathbb{RP}^1$.*

Proof. It is enough to work in the affine chart $[-t : 1]$, $t \in \mathbb{R}$; the single missing projective direction can be added at the end. Put

$$B_t = A_2 - tA_1, \quad b_t = a_2 - ta_1, \quad K_t = \ker B_t.$$

For every t whose stationary set is nonempty, choose one x_t satisfying $B_t x_t + b_t = 0$ and put $g_t = A_1 x_t + a_1$.

If $h \in K_t$, exact Taylor expansion gives

$$f_1(x_t + h) - f_1(x_t) = 2\langle g_t, h \rangle + \langle A_1 h, h \rangle. \quad (4)$$

The corresponding change in f_2 is t times the right side, because $f_2 - t f_1$ is constant on the affine stationary set $x_t + K_t$.

First consider the set Q of parameters t for which the restriction of the symmetric bilinear form $\langle A_1 \cdot, \cdot \rangle$ to K_t is nonzero. For every $t \in Q$, choose $h_t, k_t \in K_t$ with $\langle A_1 h_t, k_t \rangle \neq 0$. If $s \neq t$ and $k_s \in K_s$, then

$$t\langle A_1 h_t, k_s \rangle = \langle A_2 h_t, k_s \rangle = \langle h_t, A_2 k_s \rangle = s\langle A_1 h_t, k_s \rangle.$$

Thus $\langle A_1 h_t, k_s \rangle = 0$. After rescaling, $(A_1 h_t, k_t)_{t \in Q}$ is a biorthogonal system, so Lemma 2 makes Q countable.

Now let $t \notin Q$. The quadratic term in (4) vanishes identically on K_t , so the fiber is active exactly when $\langle g_t, \cdot \rangle|_{K_t}$ is nonzero. Let L be this set of parameters. If $s \neq t$ and $h_s \in K_s$, stationarity at t gives

$$\langle A_2 x_t + a_2, h_s \rangle = t\langle A_1 x_t + a_1, h_s \rangle.$$

Self-adjointness and $A_2 h_s = s A_1 h_s$ give

$$\langle A_2 x_t + a_2, h_s \rangle = s\langle A_1 x_t + a_1, h_s \rangle;$$

here we also used $\langle a_2 - s a_1, h_s \rangle = 0$, obtained by pairing $B_s x_s + b_s = 0$ with h_s . Hence

$$\langle g_t, h_s \rangle = 0, \quad s \neq t.$$

For each $t \in L$, choose $v_t \in K_t$ with $\langle g_t, v_t \rangle \neq 0$. After rescaling, $(g_t, v_t)_{t \in L}$ is biorthogonal. Therefore L is countable.

By (4), every active parameter belongs to $Q \cup L$. Adding the missing projective direction proves the lemma. $\square$

3.5 The area estimate

Let $\mathcal{E}$ be the countable set of active directions from Lemma 3. Fix $\eta > 0$ and cover $\mathcal{E}$ by an open set $O_\eta \subset P$ whose component arcs J_s have total length less than η . We may take η small enough that all these arcs admit aligned representatives. The set $f(B_H(R))$ is bounded, say by M_R . By (3), the values with multiplier in J_s lie in the intersection of a ball of radius M_R with a strip of width $2C'_R|J_s|$. Hence their planar outer measure is at most $C'_R|J_s|$. Countable subadditivity gives

$$\mathcal{L}^{2*}(f(\mathcal{I}_R|_{O_\eta})) \leq C'_R\eta. \quad (5)$$

On the compact complement $F_\eta = P \setminus O_\eta$, the stationary point need not be unique, but its value under f is unique. The projection of $\mathcal{I}_R|_{F_\eta}$ onto its compact multiplier image is a continuous surjection from a compact space to a Hausdorff space and hence a quotient map. Since f is constant on every projection fiber, it induces a continuous selected value

$$y([\lambda]) = f(x_\lambda)$$

which is uniformly continuous on the compact multiplier set where it is defined. Let $\omega_\eta(\delta) \rightarrow 0$ be a modulus of continuity.

Partition P into $O(\delta^{-1})$ arcs of length at most δ . On each arc, the selected values have diameter at most $\omega_\eta(\delta)$ and, by (3), normal width at most $2C'_R\delta$. They therefore fit in a rectangle of area at most $C''_R\delta\omega_\eta(\delta)$. Summing over the arcs and letting $\delta \downarrow 0$ gives

$$\mathcal{L}^2(f(\mathcal{I}_R|_{F_\eta})) = 0.$$

Together with (5), this shows

$$\mathcal{L}^2(f(\text{Crit}(f) \cap B_H(R))) \leq C'_R\eta.$$

Let $\eta \downarrow 0$. The bounded critical-value set is null. Finally, $\text{Crit}(f) = \bigcup_{N \geq 1} (\text{Crit}(f) \cap B_H(N))$, so countable subadditivity proves Theorem 1.

4 Limitations and verification

The source expects global Sard for all $f \in \mathcal{P}_2^m(H)$. The present argument proves the expectation completely for $m = 2$, including noncommuting pencils with uncountably many singular projective directions. It makes no claim for $m > 2$. The biorthogonality step is special to a two-dimensional target: two distinct projective multipliers force the relevant mixed pairing to vanish, whereas in higher target dimension it need only lie in the intersection of two hyperplanes.

The Taylor identity was independently checked by the included numerical script on random finite-dimensional symmetric pencils. This check is not part of the proof. The bounded literature search described in the README found no exact statement of the unrestricted target-dimension-two theorem; novelty confidence is moderate.

References

- [1] A. Lerario, L. Rizzi, and D. Tiberio, *Sard properties for polynomial maps in infinite dimension*, arXiv:2407.02296v4, accepted in the Journal of the European Mathematical Society.
- [2] J. B. Conway, *A Course in Functional Analysis*, 2nd ed., Springer, 1990.